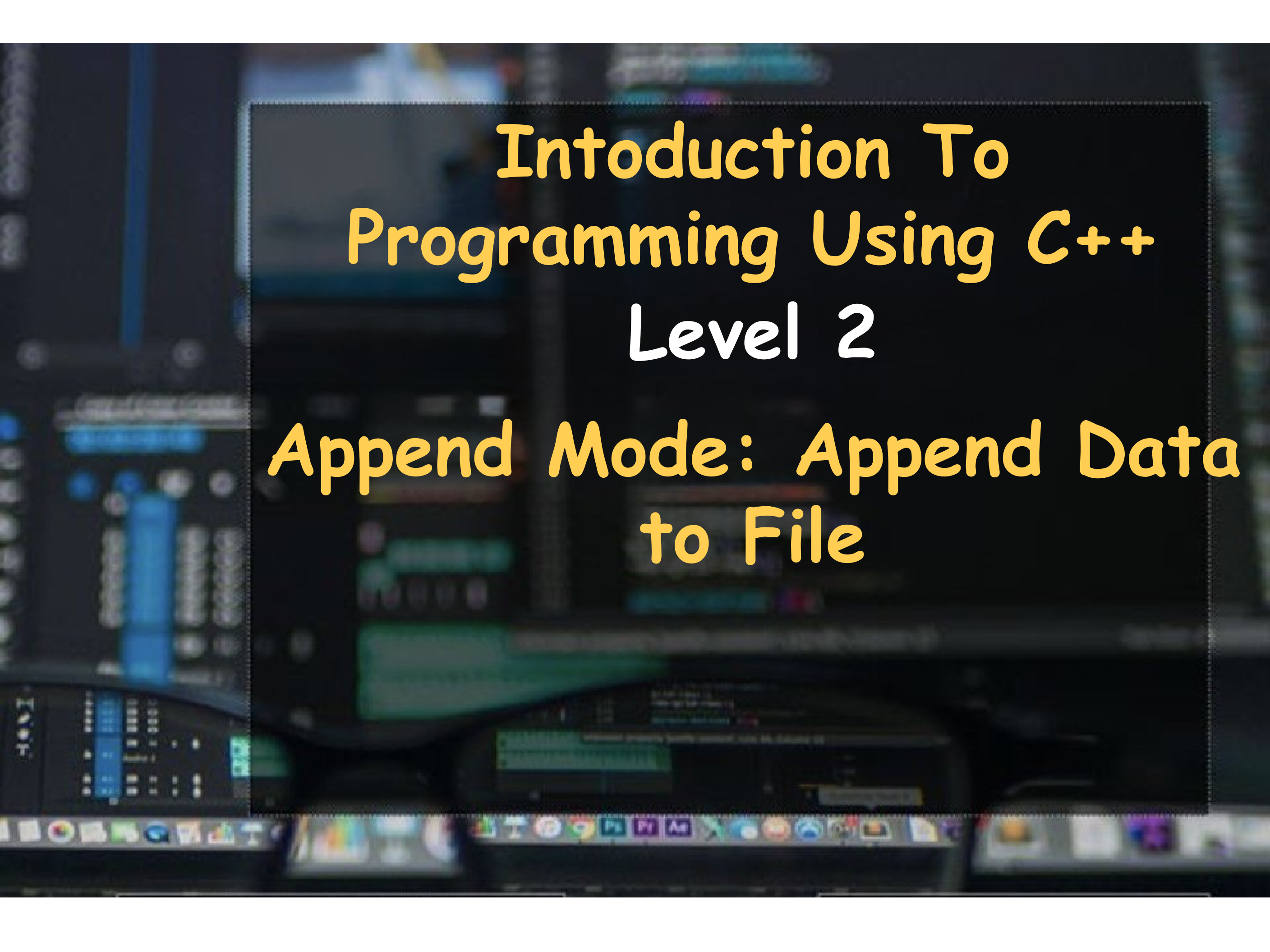
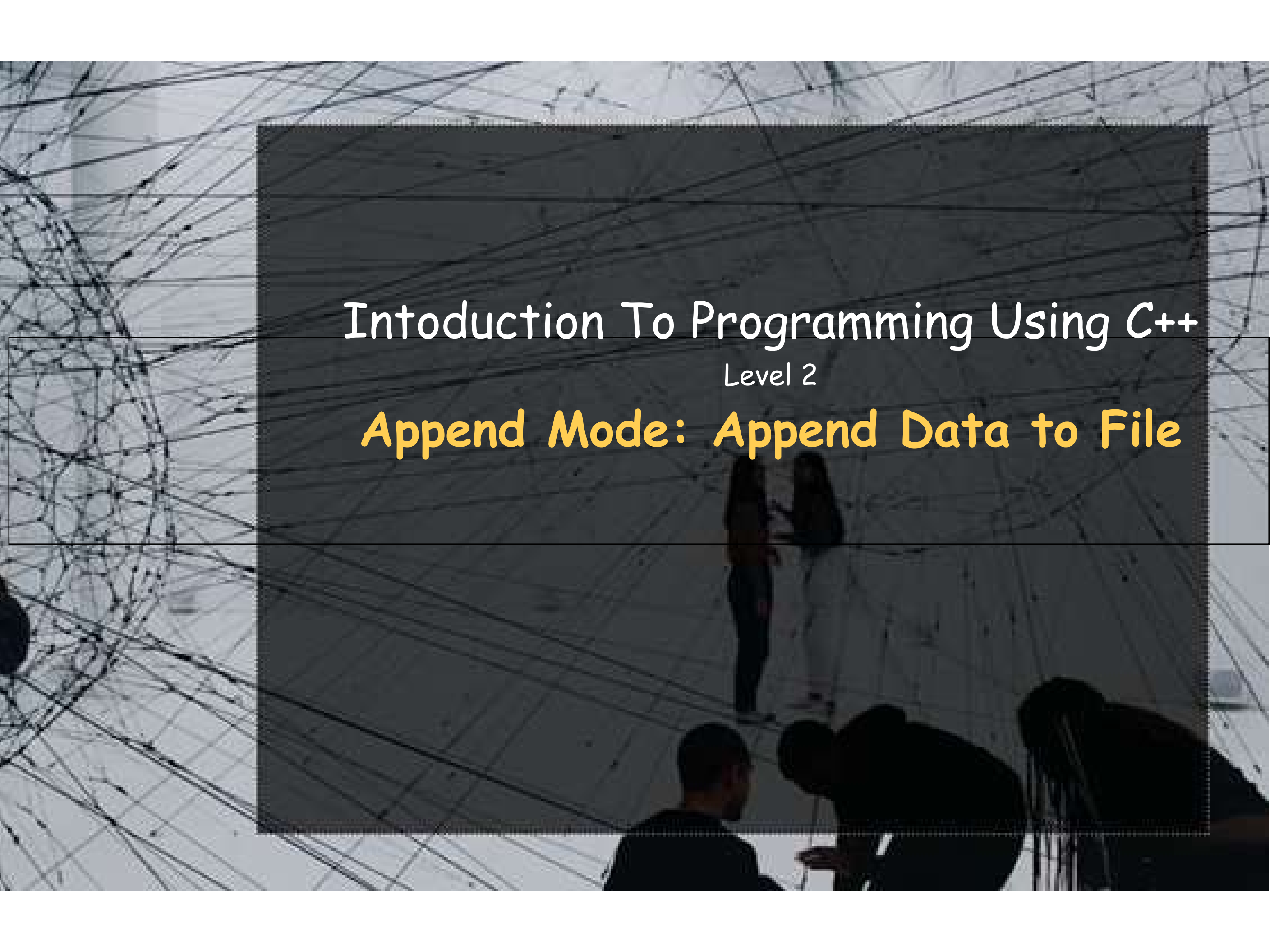


The background is a blurred image of a computer monitor. The screen shows a dark-themed code editor with some text in green and blue. At the bottom, a Windows taskbar is visible with various application icons like Chrome, Photoshop, and others. The text is overlaid on this background.

Introduction To Programming Using C++ Level 2

Append Mode: Append Data to File



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Level 2

Append Mode: Append Data to File

Definition

Append Mode

 Append Mode in C++ allows new data to be added to the end of a file without removing the existing data.

 It is useful when we want to keep previous data and continuously add new data to the same file.

Explanation

`ios::app`

-  `ios::app` opens the file in append mode.
-  New data is written at the end of the file.
-  Existing data is preserved.

```
file.open("PATH/file.extension", ios::app);
```

Combining `ios::out` and `ios::app`

-  We can combine `ios::out` and `ios::app` using the bitwise OR operator `|`.

```
file.open("PATH/file.extension", ios::out | ios::app);
```

-  `ios::out` enables output/writing.
-  `ios::app` makes every write operation occur at the end of the file.
-  The file is created if it does not already exist.

Append Mode vs Write Mode

-  `ios::out` is commonly used when writing a new version of the file and may truncate existing contents.
-  `ios::app` preserves the existing contents and adds new data at the end.

Procedures

1. Create an `fstream` object.
2. Open the file using `ios::app` or `ios::out | ios::app`.
3. Check whether the file opened successfully.
4. Write the new data using `<<`.
5. The new data is added to the end of the file.
6. Close the file using `close()`.

💡 Example

```
int main()
{
    fstream MyFile;
    MyFile.open("MyFile.txt", ios::out | ios::app);    // Append Mode

    if (MyFile.is_open())
    {
        MyFile << "Hi, this is a new line\n";
        MyFile << "Hi, this is another new line\n";
        MyFile.close();    // Always close the file
    }
    return 0;
}
```

➡ Result

If the file already contains:

Hi, this is the first line

Hi, this is the second line

After running the program, the new data is added to the end:

Hi, this is the first line

Hi, this is the second line

Hi, this is a new line

Hi, this is another new line

Note

- ◆ Append Mode is similar to adding a new item to the end of a list.
- ◆ Existing data is preserved, and new data is added after it.
- ◆ ``ios::app`` ensures that output is always written at the end of the file.
- ◆ ``ios::out | ios::app`` can be used when explicitly specifying output mode together with append mode.

⊘ Warning

- ◆ Do not use ``ios::out`` alone when you want to preserve existing file contents, because opening an existing file for output can truncate its contents.
- ◆ ``ios::app`` is intended for adding data to the end of the file.
- ◆ Always close the file after finishing the writing operation.



Comparison

▨ Append Mode (`ios::app`)

- ✓ Preserves existing data.
- ✓ New data is written at the end of the file.
- ✓ Useful for continuously adding new data.
- ✓ If the file does not exist, it can be created when opened for output.

▤ Write Mode (`ios::out`)

- ✓ Opens a file for output/writing.
- ✓ Existing contents may be truncated when the file is opened.
- ✓ Useful when replacing the previous file contents.

✦ Summary

- ✓ ``ios::app`` = append data without removing existing data.
- ✓ ``ios::out`` = output/write mode.
- ✓ ``ios::out | ios::app`` = explicitly combines output and append modes.
- ✓ Append mode writes new data at the end of the file.
- ✓ Existing data remains unchanged.
- ✓ The file can be created if it does not already exist.
- ✓ Close the file after writing using ``file.close();``

Thank you 😊